

Eyad Salama

eyad.salama@mila.quebec | esalama057@proton.me | linkedin.com/in/esalama | github.com/izarrios

RESEARCH FOCUS

Building cost-efficient AI-powered data systems by making semantic operators first-class components of relational DBMSs. Focused on cost-aware query optimization, LLM inference, execution systems, and multimodal analytics.

EXPERIENCE

Graduate Research Assistant

May 2026 – Present

Mila & DAIS Lab

Québec, Canada

- Developing cost models, query optimization, and execution techniques to reduce unnecessary model inference in scalable AI-powered data systems
- Extending Flock, a DuckDB extension for SQL-native LLM and retrieval-augmented generation workloads, by improving reliability, performance, and usability and developing semantic caching for LLM operators
- Adding asynchronous support for FFX operators with PhD collaborators in the DAIS Laboratory to enable more efficient semantic query execution

GenAI Engineer

Aug 2025 – Feb 2026

EnergyAI Berlin

Berlin, Germany

- Automated multi-step German regulatory and permitting workflows, reducing the manual work required to advance EV-charging grid-station projects toward approval
- Developed specialized agents to gather, validate, and synthesize the information required for grid-station planning and compliance
- Designed and deployed a multi-agent architecture for GridOS to automate German regulatory and permitting workflows for EV-charging grid-station development (Google ADK, Vertex Agent Engine, Vercel AI SDK)
- Built Databricks pipelines to create validated training data from web and structured sources, enabling model fine-tuning for energy-grid operations

Applied Scientist

Jul 2024 – Aug 2025

Microsoft

Cairo, Egypt

- Worked on the NLP inference engine powering SwiftKey, as well as Windows, Samsung keyboard and other partners (C++)
- Adapted the engine for GPT-2-based models, extending support to modern transformer architectures
- Contributed in the implementation of the ghost typing feature on the engine
- Conducted benchmarking of engine and model integration (quality and performance) across multiple platforms, identifying and resolving critical performance bottlenecks, resulting in a 15% reduction in inference latency.
- Developed a tool for evaluating and optimizing engine by searching with various strategies and hyperparameters to assess inference engine performance and accuracy, improving en_US locale predictions accuracy by 5%.

Embedded Software Engineer

Sep 2023 – Jul 2024

Siemens

Cairo, Egypt

- Developed and integrated a CAN communication stack compliant with AUTOSAR standards, including message parsing, transmission scheduling, and error handling (C/C++)
- Implemented a bootloader in Embedded C with Firmware Over-The-Air (FOTA), reducing update deployment time by 50%.
- Designed and deployed real-time tasks on OSEK/VDX OS, optimizing system performance and reliability, leading to a 10% improvement in task execution efficiency.

Software Engineer

Oct 2023 – Jul 2024

ITTIA

Bellevue, USA

- Implemented database optimization algorithms resulting in 30% query performance improvement (C/C++)
- Led the development of a new UI tool used for users to interact with dashboards which included multiple system/database analytics, increasing user engagement by 25% (React, Typescript)
- Increased test coverage by 30% across multiple components of the product.

- Identified and resolved several critical performance bottlenecks on various embedded devices and platforms, such as optimizing memory usage on ARM Cortex-M microcontrollers and reducing boot times on IoT gateways.

High Performance Computing Intern

Jul 2023 – Sep 2023

Brightskies

Alexandria, Egypt

- Worked on an internal seismic data processor in supporting multiple industry formats, supporting parallel I/O and CUDA computations for large-scale geophysical datasets (C++, CUDA)
- Developed a user-friendly command-line interface to encapsulate and simplify usage of an internal library, improving team productivity

EDUCATION

Polytechnique Montréal

Montréal, QC, Canada

Master of Science, Computer Engineering

May 2026 – Present

Alexandria University

Alexandria, Egypt

Bachelor of Science, Computer and Communication Engineering

Sep 2019 – Jul 2024

OPEN SOURCE CONTRIBUTIONS

- [AlexsJones/llmfit](#)
- [dais-polymtl/flock](#)
- [Chatterino/chatterino2](#)
- [yorukot/superfile](#)

CERTIFICATIONS

Databricks Certified Machine Learning Professional | Databricks Certified Machine Learning Associate | Databricks Certified Generative AI Engineer Associate | Databricks Certified Data Engineer Associate

Digital Egypt Pioneers Programs (DEPI) | Developing Secure Software (The Linux Foundation) | Machine Learning Specialization (Coursera)

Advanced Learning Algorithms (Coursera) | Supervised Machine Learning: Regression and Classification (Coursera)

Unsupervised Learning, Recommenders, Reinforcement Learning (Coursera) | AWS Academy Cloud Foundations

SKILLS

C++, PyTorch, Machine learning, Artificial intelligence, CUDA, Triton, Python, Go, Rust, C, OMP, ONNX Runtime, MPI, Make, CMake, Ninja, Scons, Java, Scala, Docker, Git, Typescript, React